

Workshop Module – Development & QA Task Allocation (Keerthana & Neha) – 5-Day Execution Plan

Developer - Day 1 (Highest Priority)(Keerthana)

Attendance Automation & Session Lifecycle Stabilization

Objective

Complete the entire attendance workflow and session lifecycle synchronization.

Tasks

1. Attendance Automation

- Fix MongoDB duplicate attendance (E11000) issue completely.
- Ensure attendance record is created only once per trainee per session.
- Prevent duplicate attendance creation.
- Validate workshopId, sessionId, batchId, studentId before insert.
- Store:
 - Workshop ID
 - Batch ID
 - Session ID
 - Student ID
 - Trainer ID
 - Join Time
 - Leave Time
 - Attendance Duration
 - Attendance Percentage
 - Status (Present/Late/Absent/Partial)
 - Device
 - Browser
 - IP Address (optional)

2. Attendance Auto Update

Automatically update attendance:

- Trainer Dashboard
- Admin Dashboard
- Trainee Dashboard
- Workshop Attendance Page

No manual refresh.

Use Socket.IO.

3. Session Lifecycle

Implement automatic state transitions:

Draft



Published



Scheduled



Waiting for Trainer



Live



Completed



Attendance Locked



Feedback Open



Certificate Eligible



Archived

4. Join Rules

Before start:

- Countdown
- Join disabled

Trainer started:

- Join enabled automatically

Trainer ended:

- Remove Join button
- Show Session Completed

5. Dashboard Synchronization

Synchronize:

- Live participant count
- Attendance
- Session status
- Dashboard widgets

Deliverables

- Attendance working end-to-end
 - Admin attendance updated
 - Trainer attendance updated
 - Trainee attendance updated
 - Session completion synchronized
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QA - Day 1(Neha)

Verify

- Registration
- Approval
- Batch Assignment
- Session Creation
- Session Scheduling
- Trainer Start Session
- Trainee Join
- Duplicate attendance not created
- Attendance automatically appears
- Join Time correct
- Leave Time correct
- Attendance % correct

- Present/Late/Absent correct
- Session End updates everywhere
- Join button removed after session ends

Test all edge cases.

Developer - Day 2(Keerthana)

Recording Module (Production)

Objective

Complete LiveKit Recording.

Tasks

Recording Controls

Trainer only:

- Start Recording
- Pause Recording
- Resume Recording
- Stop Recording

Recording Status

Show

- Recording indicator
- Duration
- Status
- Progress

Backend

Integrate

- LiveKit Egress
- Recording APIs
- MongoDB metadata

Store

- Recording Name
- Session
- Workshop
- Trainer
- Duration
- File Size
- Recording URL
- Thumbnail
- Status

Recording Management

Admin

Workshop



Recordings

Trainer

Workshop



Recordings

Features

- Search
- Filter
- Download
- Publish
- Delete

Students

Only published recordings visible.

Deliverables

Complete recording workflow.

QA - Day 2(Neha)

Verify

- Recording starts
 - Recording pauses
 - Resume works
 - Stop works
 - File generated
 - Recording saved in DB
 - Recording visible
 - Download works
 - Publish works
 - Student visibility works
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Developer - Day 3(Keerthana)

Feedback Module

Objective

Complete feedback workflow.

Tasks

After session completion

Trainee Dashboard



Feedback button



Feedback Form

Fields

- Overall Rating
- Trainer Rating
- Content Rating
- Audio Rating

- Video Rating
- Comments
- Suggestions

Rules

- One submission only
- Cannot edit after submission

Trainer Dashboard

Show

- Average Rating
- Individual Feedback
- Analytics
- Charts

Admin Dashboard

Show

- Overall analytics
- Filters
- Export

Socket.IO updates.

QA - Day 3(Neha)

Verify

- Feedback button appears
 - Hidden before completion
 - Submit once only
 - Validation works
 - Trainer receives feedback
 - Admin receives feedback
 - Charts updated
 - Analytics correct
-

Developer - Day 4(Keerthana)

Documents, Resources & Certificate Module

Tasks

Workshop Resources

Trainer Upload

- PDF
- PPT
- DOC
- DOCX
- XLS
- ZIP
- Images
- Videos

Students

- View
- Download

Admin

- Manage resources

Certificates

Generate only if

- Attendance $\geq$ required
- Feedback submitted
- Workshop completed

Generate PDF

Store

- Certificate Number
- Student
- Workshop
- Generated Date

Download

Verify duplicate prevention.

QA - Day 4(Neha)

Verify

Resource Upload

- PDF
- PPT
- Video
- Image

Download

Permission checks

Certificate eligibility

Certificate download

Certificate data

Developer - Day 5(Keerthana)

Production Stabilization & Final QA Fixes

Tasks

Socket.IO

Realtime updates

Performance

Error handling

Authorization

Validation

Security

Dashboard synchronization

Reports

Export

CSV

Excel

PDF

Cleanup

Remove mock data

Remove unused APIs

Code optimization

Fix console errors

Fix warnings

Final E2E testing

Complete documentation

QA - Day 5(Neha)

Complete End-to-End Testing

Admin



Create Workshop



Registration



Approval



Assign Batch



Assign Trainer



Create Session



Schedule Session



Trainer Login



Start Session



Recording Start



Trainee Join



Attendance



Announcements



Resources



Recording Stop



End Session



Attendance Locked



Feedback Enabled



Feedback Submitted



Certificate Generated



Reports Updated



Recording Available



Admin Analytics Updated

Final Deliverables Before Production

Developer (Keerthana)

- ☒ Attendance Automation
- ☒ Recording Module
- ☒ Recording Management
- ☒ Feedback Workflow
- ☒ Resource Upload Module
- ☒ Certificate Generation
- ☒ Dashboard Synchronization
- ☒ Session Lifecycle
- ☒ Socket.IO Realtime Updates
- ☒ Reports & Export
- ☒ Security Validation
- ☒ Production Bug Fixes
- ☒ Performance Optimization
- ☒ Code Cleanup
- ☒ Technical Documentation

QA(Neha)

- ☒ Attendance Testing
- ☒ Live Session Testing
- ☒ Recording Testing
- ☒ Feedback Testing
- ☒ Resource Upload Testing
- ☒ Certificate Testing
- ☒ Dashboard Synchronization Testing
- ☒ Session Lifecycle Testing
- ☒ Security & Authorization Testing
- ☒ Regression Testing
- ☒ Cross-browser Testing
- ☒ End-to-End Production Validation